

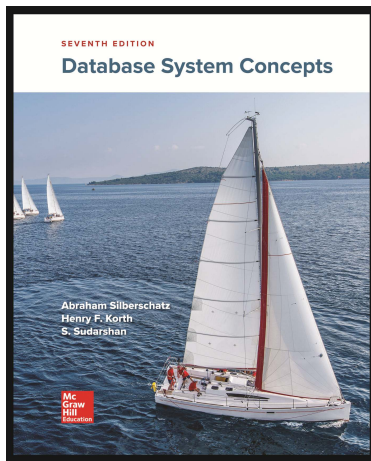
Database Administration

Lecture 09: Data Analytics.

Silberschatz, Korth & Sudarshan

5 de octubre de 2026

Database Administration: Data Analytics.



Content has been extracted from *Database System Concepts 7ed (Chapter 11)*, by Abraham Silberschatz, Henry Korth & S. Sudarshan, 2019. Visit <https://db-book.com/>.

Plan

Overview

Data Warehousing

OLAP

Data Mining



Overview

- **Data analytics:** the processing of data to infer patterns, correlations, or models for prediction
- Primarily used to make business decisions
 - Per individual customer
 - E.g. what product to suggest for purchase
 - Across all customers
 - E.g. what products to manufacture/stock, in what quantity
- Critical for businesses today



Overview (Cont.)

- Common steps in data analytics
 - Gather data from multiple sources into one location
 - Data warehouses also integrated data into common schema
 - Data often needs to be **extracted** from source formats, **transformed** to common schema, and **loaded** into the data warehouse
 - Can be done as **ETL (extract-transform-load)**, or **ELT (extract-load-transform)**
 - Generate aggregates and reports summarizing data
 - Dashboards showing graphical charts/reports
 - **Online analytical processing (OLAP) systems** allow interactive querying
 - Statistical analysis using tools such as R/SAS/SPSS
 - Including extensions for parallel processing of big data
 - Build **predictive models** and use the models for decision making



Overview (Cont.)

- Predictive models are widely used today
 - E.g. use customer profile features (e.g. income, age, gender, education, employment) and past history of a customer to predict likelihood of default on loan
 - and use prediction to make loan decision
 - E.g. use past history of sales (by season) to predict future sales
 - And use it to decide what/how much to produce/stock
 - And to target customers
- Other examples of business decisions:
 - What items to stock?
 - What insurance premium to change?
 - To whom to send advertisements?



Overview (Cont.)

- **Machine learning** techniques are key to finding patterns in data and making predictions
- **Data mining** extends techniques developed by machine-learning communities to run them on very large datasets
- The term **business intelligence (BI)** is synonym for data analytics
- The term **decision support** focuses on reporting and aggregation

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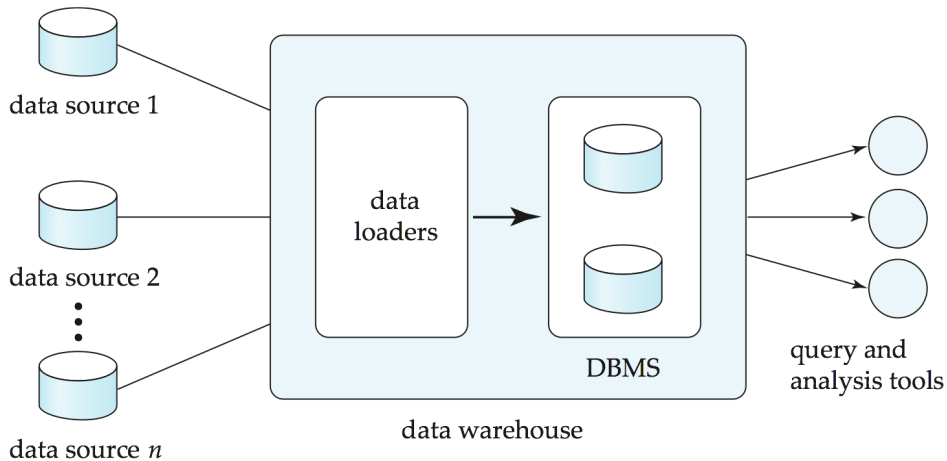


Data Warehousing

- Data sources often store only current data, not historical data
- Corporate decision making requires a unified view of all organizational data, including historical data
- A **data warehouse** is a repository (archive) of information gathered from multiple sources, stored under a unified schema, at a single site
 - Greatly simplifies querying, permits study of historical trends
 - Shifts decision support query load away from transaction processing systems



Data Warehousing





Design Issues

- *When and how to gather data*
 - **Source driven architecture:** data sources transmit new information to warehouse
 - either continuously or periodically (e.g. at night)
 - **Destination driven architecture:** warehouse periodically requests new information from data sources
 - **Synchronous vs asynchronous replication**
 - Keeping warehouse exactly synchronized with data sources (e.g. using two-phase commit) is often too expensive
 - Usually OK to have slightly out-of-date data at warehouse
 - Data/updates are periodically downloaded from online transaction processing (OLTP) systems.
- *What schema to use*
 - Schema integration



More Warehouse Design Issues

- **Data transformation** and **data cleansing**
 - E.g. correct mistakes in addresses (misspellings, zip code errors)
 - **Merge** address lists from different sources and **purge** duplicates
- *How to propagate updates*
 - Warehouse schema may be a (materialized) view of schema from data sources
 - View maintenance
- *What data to summarize*
 - Raw data may be too large to store on-line
 - Aggregate values (totals/subtotals) often suffice
 - Queries on raw data can often be transformed by query optimizer to use aggregate values

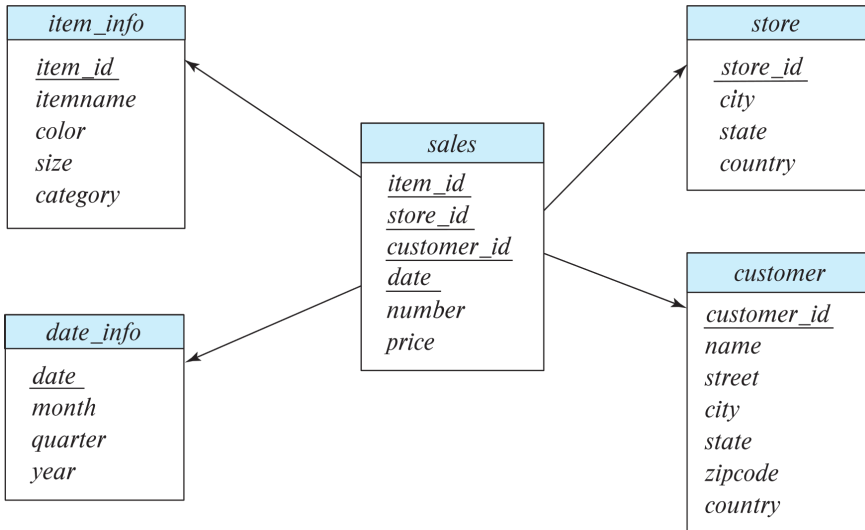


Multidimensional Data and Warehouse Schemas

- Data in warehouses can usually be divided into
 - **Fact tables**, which are large
 - E.g *sales(item_id, store_id, customer_id, date, number, price)*
 - **Dimension tables**, which are relatively small
 - Store extra information about stores, items, etc
- Attributes of fact tables can be usually viewed as
 - **Measure attributes**
 - measure some value, and can be aggregated upon
 - e.g., the attributes *number* or *price* of the *sales* relation
 - **Dimension attributes**
 - dimensions on which measure attributes are viewed
 - e.g., attributes *item_id*, *color*, and *size* of the *sales* relation
 - Usually small ids that are foreign keys to dimension tables



Data Warehouse Schema





Multidimensional Data and Warehouse Schemas

- Resultant schema is called a **star schema**
 - More complicated schema structures
 - **Snowflake schema**: multiple levels of dimension tables
 - **May have** multiple fact tables
- Typically
 - fact table joined with dimension tables and then
 - group-by on dimension table attributes, and then
 - aggregation on measure attributes of fact table
- Some applications do not find it worthwhile to bring data to a common schema
 - **Data lakes** are repositories which allow data to be stored in multiple formats, without schema integration
 - Less upfront effort, but more effort during querying



Database Support for Data Warehouses

- Data in warehouses usually append only, not updated
 - Can avoid concurrency control overheads
- Data warehouses often use **column-oriented storage**
 - E.g. a sequence of *sales* tuples is stored as follows
 - Values of *item_id* attribute are stored as an array
 - Values of *store_id* attribute are stored as an array,
 - And so on
 - Arrays are compressed, reducing storage, IO and memory costs significantly
 - Queries can fetch only attributes that they care about, reducing IO and memory cost
 - More details in Section 13.6
- Data warehouses often use parallel storage and query processing infrastructure
 - Distributed file systems, Map-Reduce, Hive, ...

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Data Analysis and OLAP

- **Online Analytical Processing (OLAP)**

- Interactive analysis of data, allowing data to be summarized and viewed in different ways in an online fashion (with negligible delay)

- We use the following relation to illustrate OLAP concepts

- *sales (item_name, color, clothes_size, quantity)*

This is a simplified version of the *sales* fact table joined with the dimension tables, and many attributes removed (and some renamed)



Example sales relation

<i>item_name</i>	<i>color</i>	<i>clothes_size</i>	<i>quantity</i>
dress	dark	small	2
dress	dark	medium	6
dress	dark	large	12
dress	pastel	small	4
dress	pastel	medium	3
dress	pastel	large	3
dress	white	small	2
dress	white	medium	3
dress	white	large	0
pants	dark	small	14
pants	dark	medium	6
pants	dark	large	0
pants	pastel	small	1
pants	pastel	medium	0
pants	pastel	large	1
pants	white	small	3
pants	white	medium	0
pants	white	large	2
shirt	dark	small	2
shirt	dark	medium	6
shirt	dark	large	6
shirt	pastel	small	4
shirt	pastel	medium	1
shirt	pastel	large	2
shirt	white	small	17
shirt	white	medium	1
shirt	white	large	10
skirt	dark	small	2
skirt	dark	medium	5
...	...	...	...
...	...	...	...



Cross Tabulation of sales by *item_name* and *color*

clothes_size all

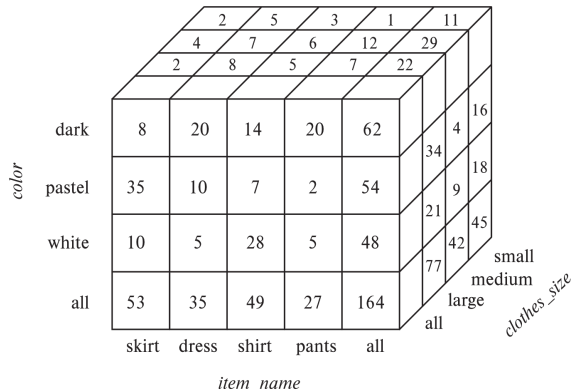
		<i>color</i>			
		dark	pastel	white	total
<i>item_name</i>	skirt	8	35	10	53
	dress	20	10	5	35
	shirt	14	7	28	49
	pants	20	2	5	27
	total	62	54	48	164

- The table above is an example of a **cross-tabulation** (**cross-tab**), also referred to as a **pivot-table**.
 - Values for one of the dimension attributes form the row headers
 - Values for another dimension attribute form the column headers
 - Other dimension attributes are listed on top
 - Values in individual cells are (aggregates of) the values of the dimension attributes that specify the cell.



Data Cube

- A **data cube** is a multidimensional generalization of a cross-tab
- Can have n dimensions; we show 3 below
- Cross-tabs can be used as views on a data cube





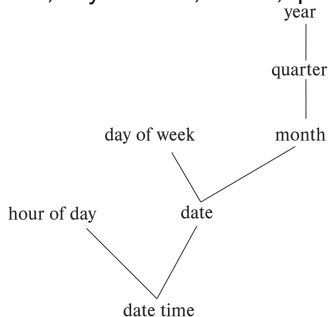
Online Analytical Processing Operations

- **Pivoting:** changing the dimensions used in a cross-tab
 - E.g. moving colors to column names
- **Slicing:** creating a cross-tab for fixed values only
 - E.g. fixing color to white and size to small
 - Sometimes called **dicing**, particularly when values for multiple dimensions are fixed.
- **Rollup:** moving from finer-granularity data to a coarser granularity
 - E.g. aggregating away an attribute
 - E.g. moving from aggregates by day to aggregates by month or year
- **Drill down:** The opposite operation - that of moving from coarser-granularity data to finer-granularity data



Hierarchies on Dimensions

- **Hierarchy** on dimension attributes: lets dimensions be viewed at different levels of detail
- E.g., the dimension *datetime* can be used to aggregate by hour of day, date, day of week, month, quarter or year



(a) time hierarchy



(b) location hierarchy



Cross Tabulation With Hierarchy

- Cross-tabs can be easily extended to deal with hierarchies
- Can drill down or roll up on a hierarchy
- E.g. hierarchy: *item_name* → *category*

clothes_size:

all

<i>category</i>	<i>item_name</i>	<i>color</i>				
		dark	pastel	white	total	
womenswear	skirt	8	8	10	53	88
	dress	20	20	5	35	
	subtotal	28	28	15		
menswear	pants	14	14	28	49	76
	shirt	20	20	5	27	
	subtotal	34	34	33		
total		62	62	48		164



Relational Representation of Cross-tabs

- Cross-tabs can be represented as relations
- We use the value **all** to represent aggregates.
- The SQL standard actually uses *null* values in place of **all**
 - Works with any data type
 - But can cause confusion with regular null values.

<i>item_name</i>	<i>color</i>	<i>clothes_size</i>	<i>quantity</i>
skirt	dark	all	8
skirt	pastel	all	35
skirt	white	all	10
skirt	all	all	53
dress	dark	all	20
dress	pastel	all	10
dress	white	all	5
dress	all	all	35
shirt	dark	all	14
shirt	pastel	all	7
shirt	white	all	28
shirt	all	all	49
pants	dark	all	20
pants	pastel	all	2
pants	white	all	5
pants	all	all	27
all	dark	all	62
all	pastel	all	54
all	white	all	48
all	all	all	164

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Data Mining

- **Data mining** is the process of semi-automatically analyzing large databases to find useful patterns
 - Similar goals to machine learning, but on very large volumes of data
- Part of the larger area of **knowledge discovery in databases (KDD)**
- Some types of knowledge can be represented as rules
- More generally, knowledge is discovered by applying machine learning techniques on past instances of data, to form a **model**
 - Model is then used to make predictions for new instances



Types of Data Mining Tasks

- **Prediction** based on past history
 - Predict if a credit card applicant poses a good credit risk, based on some attributes (income, job type, age, ..) and past history
 - Predict if a pattern of phone calling card usage is likely to be fraudulent
- Some examples of prediction mechanisms:
 - **Classification**
 - Items (with associated attributes) belong to one of several classes
 - **Training instances** have attribute values and classes provided
 - Given a new item whose class is unknown, predict to which class it belongs based on its attribute values
 - **Regression** formulae
 - Given a set of mappings for an unknown function, predict the function result for a new parameter value



Data Mining (Cont.)

■ Descriptive Patterns

• Associations

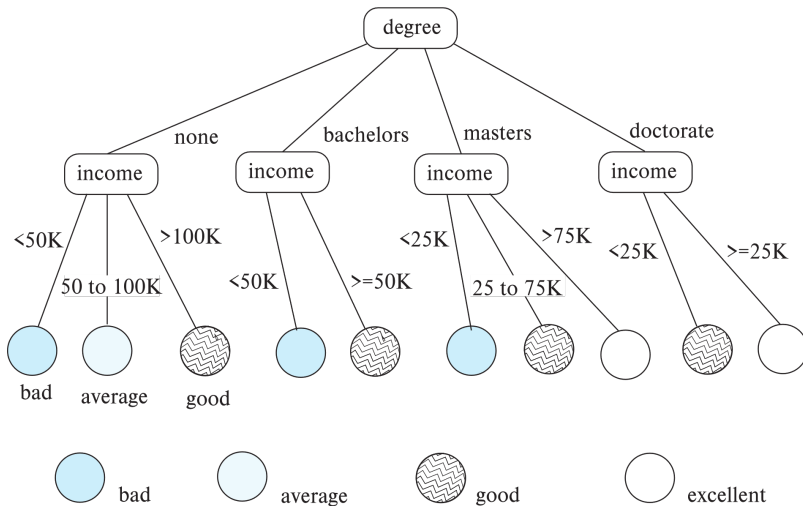
- Find books that are often bought by “similar” customers. If a new such customer buys one such book, suggest the others too.
- Associations may be used as a first step in detecting **causation**
 - E.g. association between exposure to chemical X and cancer,

• Clusters

- E.g. typhoid cases were clustered in an area surrounding a contaminated well
- Detection of clusters remains important in detecting epidemics



Decision Tree Classifiers





Decision Trees

- Each internal node of the tree partitions the data into groups based on a **partitioning attribute**, and a **partitioning condition** for the node
- Leaf node:
 - all (or most) of the items at the node belong to the same class, or
 - all attributes have been considered, and no further partitioning is possible.
- Traverse tree from top to make a prediction
- Number of techniques for constructing decision tree classifiers
 - We omit details



Bayesian Classifiers

- Bayesian classifiers use **Bayes theorem**, which says

$$p(c_j | d) = \frac{p(d | c_j) p(c_j)}{p(d)}$$

where

$p(c_j | d)$ = probability of instance d being in class c_j ,

$p(d | c_j)$ = probability of generating instance d given class c_j ,

$p(c_j)$ = probability of occurrence of class c_j , and

$p(d)$ = probability of instance d occurring



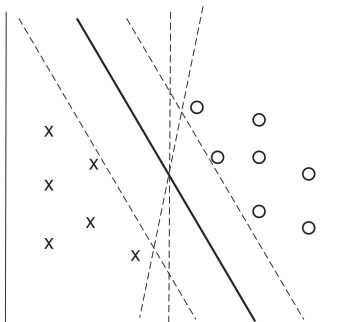
Naïve Bayesian Classifiers

- Bayesian classifiers require
 - computation of $p(d | c_j)$
 - precomputation of $p(c_j)$
 - $p(d)$ can be ignored since it is the same for all classes
- To simplify the task, **naïve Bayesian classifiers** assume attributes have independent distributions, and thereby estimate
$$p(d | c_j) = p(d_1 | c_j) * p(d_2 | c_j) * \dots * (p(d_n | c_j))$$
 - Each of the $p(d_i | c_j)$ can be estimated from a histogram on d_i values for each class c_j
 - the histogram is computed from the training instances
 - Histograms on multiple attributes are more expensive to compute and store



Support Vector Machine Classifiers

- Simple 2-dimensional example:
 - Points are in two classes
 - Find a line (**maximum margin line**) s.t. line divides two classes, and distance from nearest point in either class is maximum





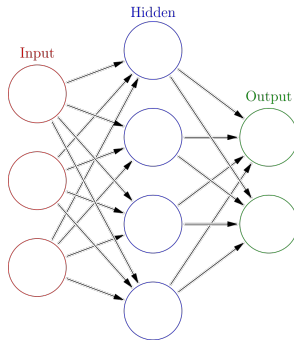
Support Vector Machine

- In n -dimensions points are divided by a plane, instead of a line
- SVMs can be used separators that are curve, not necessarily linear, by transforming points before classification
 - Transformation functions may be non-linear and are called kernel functions
 - Separator is a plane in the transformed space, but maps to curve in original space
- There may not be an exact planar separator for a given set of points
 - Choose plane that best separates points
- N -ary classification can be done by N binary classifications
 - In class i vs. not in class i .



Neural Network Classifiers

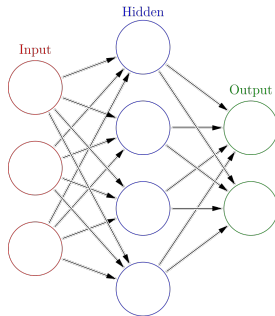
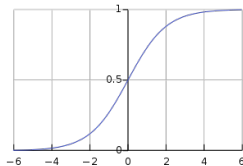
- Neural network has multiple layers
 - Each layer acts as input to next later
- First layer has input nodes, which are assigned values from input attributes
- Each node combines values of its inputs using some weight function to compute its value
 - Weights are associated with edges
- For classification, each output value indicates likelihood of the input instance belonging to that class
 - Pick class with maximum likelihood
- Weights of edges are key to classification
- Edge weights are learnt during training phase





Neural Network Classifiers

- Value of a node may be linear combination of inputs, or may be a non-linear function
 - E.g. sigmoid function
- Backpropagation algorithm** works as follows
 - Weights are set randomly initially
 - Training instances are processed one at a time
 - Output is computed using current weights
 - If classification is wrong, weights are tweaked to get a higher score for the correct class





Neural Networks (Cont.)

- **Deep neural networks** have a large number of layers with large number of nodes in each layer
- **Deep learning** refers to training of deep neural network on very large numbers of training instances
- Each layer may be connected to previous layers in different ways
 - Convolutional networks used for image processing
 - More complex architectures used for text processing, and machine translation, speech recognition, etc.
- Neural networks are a large area in themselves
 - Further details beyond scope of this chapter



Regression

- Regression deals with the prediction of a value, rather than a class.
 - Given values for a set of variables, $X_1, X_2, \dots, X_n$, we wish to predict the value of a variable Y .
- One way is to infer coefficients $a_0, a_1, a_1, \dots, a_n$ such that
$$Y = a_0 + a_1 * X_1 + a_2 * X_2 + \dots + a_n * X_n$$
- Finding such a linear polynomial is called **linear regression**.
 - In general, the process of finding a curve that fits the data is also called **curve fitting**.
- The fit may only be approximate
 - because of noise in the data, or
 - because the relationship is not exactly a polynomial
- Regression aims to find coefficients that give the best possible fit.



Association Rules

- Retail shops are often interested in associations between different items that people buy.
 - Someone who buys bread is quite likely also to buy milk
 - A person who bought the book *Database System Concepts* is quite likely also to buy the book *Operating System Concepts*.
- Associations information can be used in several ways.
 - E.g. when a customer buys a particular book, an online shop may suggest associated books.
- **Association rules:**
 - $bread \Rightarrow milk$ $DB\text{-}Concepts, OS\text{-}Concepts \Rightarrow Networks$
 - Left hand side: **antecedent**, right hand side: **consequent**
 - An association rule must have an associated **population**; the population consists of a set of **instances**
 - E.g. each transaction (sale) at a shop is an instance, and the set of all transactions is the population



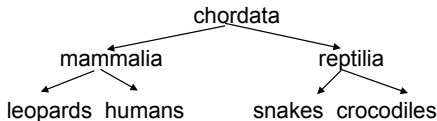
Association Rules (Cont.)

- Rules have an associated support, as well as an associated confidence.
- **Support** is a measure of what fraction of the population satisfies both the antecedent and the consequent of the rule.
 - E.g. suppose only 0.001 percent of all purchases include milk and screwdrivers. The support for the rule is $milk \Rightarrow screwdrivers$ is low.
- **Confidence** is a measure of how often the consequent is true when the antecedent is true.
 - E.g. the rule $bread \Rightarrow milk$ has a confidence of 80 percent if 80 percent of the purchases that include bread also include milk.
- We omit further details, such as how to efficiently infer association rules



Clustering

- **Clustering:** Intuitively, finding clusters of points in the given data such that similar points lie in the same cluster
- Can be formalized using distance metrics in several ways
 - Group points into k sets (for a given k) such that the average distance of points from the centroid of their assigned group is minimized
 - Centroid: point defined by taking average of coordinates in each dimension.
 - Another metric: minimize average distance between every pair of points in a cluster
- **Hierarchical clustering:** example from biological classification
 - (the word classification here does not mean a prediction mechanism)





Clustering and Collaborative Filtering

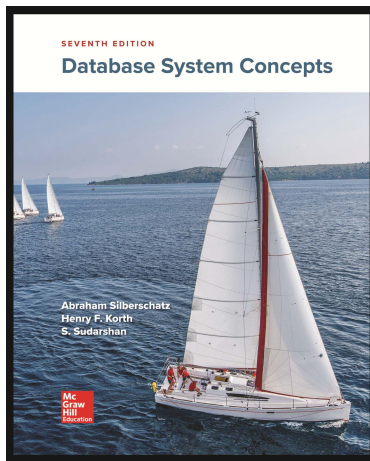
- Goal: predict what movies/books/... a person may be interested in, on the basis of
 - Past preferences of the person
 - Preferences of other people
- One approach based on repeated clustering
 - Cluster people based on their preferences for movies
 - Then cluster movies on the basis of being liked by the same clusters of people
 - Again cluster people based on their preferences for (the newly created clusters of) movies
 - Repeat above till equilibrium
 - Given new user
 - Find most similar cluster of existing users and
 - Predict movies in movie clusters popular with that user cluster
- Above problem is an instance of **collaborative filtering**



Other Types of Mining

- **Text mining**: application of data mining to textual documents
- **Sentiment analysis**
 - E.g. learn to predict if a user review is positive or negative about a product
- **Information extraction**
 - Create structured information from unstructured textual description or semi-structured data such as tabular displays
- **Entity recognition** and **disambiguation**
 - E.g. given text with name “Michael Jordan” does the name refer to the famous basketball player or the famous ML expert
- **Knowledge graph** (see Section 8.4)
 - Can be constructed by information extraction from different sources, such as Wikipedia

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